CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 23: Probability

Topic-wise short notes with solved examples

Random experiments, outcomes and events, the probability of an event with equally likely outcomes, and standard problems on coins, dice, cards and coloured balls.

1 Basic terms

- An **experiment** is an action that gives a definite result. A **random experiment** is one whose result cannot be predicted beforehand, though all the possible results are known.
- Each possible result is an **outcome**. The set of all outcomes is the **sample space** S.
- An **event** is one or more outcomes of the experiment. The outcomes that make it happen are **favourable** to it.
- Outcomes are **equally likely** when each has the same chance of happening, as with a fair coin or an unbiased die.

Experiment	Sample space	Number of outcomes
Tossing a coin	H, T	2
Tossing two coins	HH, HT, TH, TT	4
Throwing a die	1, 2, 3, 4, 5, 6	6
Drawing a card from a pack	52 cards	52

2 Probability of an event

- When all the outcomes are equally likely:

$$P(E) = \frac{\text{number of outcomes favourable to } E}{\text{total number of outcomes}}$$

- Probability is a number from 0 to 1: $0 \le P(E) \le 1$.
- An **impossible event** has probability 0. A **sure (certain) event** has probability 1.
- The event "not E" is the **complement** of E: $P(E) + P(\text{not } E) = 1$.
- The probabilities of all the outcomes of an experiment add up to 1.

Example 2.1 A coin is tossed once. Find the probability of getting a head.

SOLUTION

$\frac{1}{2}$

Example 2.2 A die is thrown once. Find the probability of getting (i) a 4 (ii) an even number (iii) a number greater than 4 (iv) a 7.

SOLUTION

Favourable outcomes: (i) 1 (ii) 2, 4, 6 (iii) 5, 6 (iv) none.

(i) $\frac{1}{6}$ **(ii)** $\frac{1}{2}$ **(iii)** $\frac{1}{3}$ **(iv) 0**

Example 2.3 The probability that it will rain tomorrow is 0.35. Find the probability that it will not rain.

SOLUTION

$1 - 0.35$

0.65

3 Coins

- Two coins tossed together (or one coin tossed twice) give 4 outcomes: HH, HT, TH, TT.
- Three coins give 8 outcomes: HHH, HHT, HTH, THH, HTT, THT, TTH, TTT.

Example 3.1 Two coins are tossed together. Find the probability of getting (i) two heads (ii) exactly one head (iii) at least one head.

SOLUTION

(i) HH (ii) HT, TH (iii) HH, HT, TH

(i) $\frac{1}{4}$ **(ii)** $\frac{1}{2}$ **(iii)** $\frac{3}{4}$

Example 3.2 Three coins are tossed. Find the probability of getting exactly two tails.

SOLUTION

Favourable: HTT, THT, TTH

$\frac{3}{8}$

4 Dice

- A die has faces 1 to 6. Prime numbers on it: 2, 3, 5. Composite numbers: 4, 6. The number 1 is neither.
- Two dice thrown together give $6 \times 6 = 36$ outcomes, written as ordered pairs (a, b).

Sum of the two dice	2	3	4	5	6	7	8	9	10	11	12
Number of ways	1	2	3	4	5	6	5	4	3	2	1

Example 4.1 A die is thrown once. Find the probability of getting a prime number.

SOLUTION

2, 3, 5

$\frac{1}{2}$

Example 4.2 Two dice are thrown together. Find the probability that (i) the sum is 8 (ii) both show the same number (iii) the sum is less than 5.

SOLUTION

(i) $(2,6),(3,5),(4,4),(5,3),(6,2)$: 5 ways. (ii) 6 doublets. (iii) Sums 2, 3, 4: $1+2+3=6$ ways.

(i) $\frac{5}{36}$ **(ii)** $\frac{1}{6}$ **(iii)** $\frac{1}{6}$

5 Playing cards

- A pack has **52 cards** in 4 suits of 13 cards each: spades and clubs (black), hearts and diamonds (red).
- Each suit has an ace, the numbers 2 to 10, and three **face cards**: jack, queen and king.

Kind of card	Number in the pack
Red cards	26
Black cards	26
Cards of one suit	13
Aces	4
Kings (or queens, or jacks)	4
Face cards	12

Example 5.1 A card is drawn at random from a well-shuffled pack. Find the probability that it is (i) a king (ii) a red card (iii) a face card (iv) the ace of spades (v) not a heart.

SOLUTION

(i) $\frac{4}{52}=\frac{1}{13}$ **(ii)** $\frac{1}{2}$ **(iii)** $\frac{12}{52}=\frac{3}{13}$ **(iv)** $\frac{1}{52}$ **(v)** $\frac{39}{52}=\frac{3}{4}$

Example 5.2 Find the probability of drawing a black queen.

SOLUTION

There are 2 black queens.

$\frac{1}{26}$

6 Balls, numbers and letters

Example 6.1 A bag has 5 red, 4 blue and 3 green balls. One ball is drawn at random. Find the probability that it is (i) blue (ii) not red (iii) red or green.

SOLUTION

Total 12 balls.

(i) $\frac{1}{3}$ **(ii)** $\frac{7}{12}$ **(iii)** $\frac{8}{12} = \frac{2}{3}$

Example 6.2 Cards numbered 1 to 20 are mixed and one is drawn. Find the probability that the number is (i) a multiple of 3 (ii) a prime (iii) a perfect square.

SOLUTION

(i) 3, 6, 9, 12, 15, 18. (ii) 2, 3, 5, 7, 11, 13, 17, 19. (iii) 1, 4, 9, 16.

(i) $\frac{3}{10}$ **(ii)** $\frac{2}{5}$ **(iii)** $\frac{1}{5}$

Example 6.3 A letter is chosen at random from the word ARITHMETIC. Find the probability that it is a vowel.

SOLUTION

The word has 10 letters; the vowels are A, I, E, I.

$\frac{2}{5}$

Example 6.4 A bag has 6 white balls and some black balls. The probability of drawing a black ball is $\frac{3}{5}$. How many black balls are there?

SOLUTION

$\frac{x}{x+6} = \frac{3}{5}$, so $5x = 3x + 18$

9

7 Experimental probability

- When an experiment is repeated many times, the **experimental probability** of an event is

$$\frac{\text{number of trials in which the event happened}}{\text{total number of trials}}$$

- As the number of trials increases, it comes closer to the theoretical probability.

Example 7.1 A coin is tossed 200 times and heads appears 112 times. Find the experimental probability of a tail.

SOLUTION

Tails: 88 times.

$$\frac{88}{200} = \frac{11}{25}$$

Example 7.2 In a batch of 500 bulbs, 20 are defective. Find the probability that a bulb chosen at random is not defective.

SOLUTION

$$\frac{480}{500} = \frac{24}{25}$$

8 Common mistakes

- Counting HT and TH as one outcome.
- Taking the number of outcomes for two dice as 12 instead of 36.
- Counting 1 as a prime number.
- Counting aces as face cards.
- Giving a probability greater than 1 or less than 0.
- For "at least one", counting only "exactly one".

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